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Gene Mutations

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Mutations

- In any species, variations can be caused
  - Changes in environment
  - Changes in heredity constitution
  - More commonly by interaction between both
- If any variation appears when environment changes and is restored when the environment is restored, then it is not inherited – Ex. Muscles – Exercise
- These are **modifications – phenotypic differences between organisms with similar genotype**

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Mutations

- Mutations are changes in the genetic makeup of an individual, which can be qualitative or quantitative
- Mutations are random in location
- The effects of mutations can be positive, negative or neutral
- Mutations can be classified variously according to
  - Type of cells
  - Size and quality
  - Origin

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Type of Cells

- **Somatic VS Gametic Mutations**
- **Somatic Mutations:**
  - Occur in body cells / non-reproductive cells
  - Genetic and evolutionary consequences are insignificant, since only single cells or their daughter cells are involved
  - However, if they occur during embryonic stage, impact is significant as large proportion of cells is involved
  - Ex. Some cancerous growths
- **Gametic Mutations:**
  - In sperms and ova
  - Heritable and immense genetic significance
  - They are the raw material for natural selection

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Size & Quality

Point & Gross Mutations

- **Point Mutations:**
  - Occur in very small segment of DNA, a single nucleotide or a pair
  - Can be classified into

Deletion	CTGGAG CTAG
Insertion	CTGGAG CTGGTGGAG
Substitution	CTGGAG CTGGGG

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Size & Quality

- Deletion mutations result in **FRAMESHIFT –** causing a wrong protein to be encoded by RNA

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- **Frameshifts** can cause great phenotypic changes
- **Gross Mutations:**
  - Involve a greater portion of a gene or entire gene
  - Can occur due to rearrangement of genes within genome

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Origin

Spontaneous and Induced Mutations

- **Spontaneous –** Occur naturally / randomly in nature and their origin is unknown
- **Induced –** Certain environmental conditions can induce mutations
  - Agents that can induce mutations – **Mutagens**
  - **Radiations –** X Rays, Gamma Rays, Ultraviolet rays
  - **Temperature**
  - **Chemicals –** Some Alkoloids, Benzene, Sodium Azide

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